



Understanding protein folding with machine learning models? The case of AlphaFold2

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Abstract

The protein folding problem (PFP) has puzzled researchers for over fifty years, with its solution promising significant scientific advances, given the critical roles proteins play in biological functions. DeepMind's AlphaFold2 (AF2) has dramatically advanced the field by accurately predicting 215 million protein structures from diverse organisms – far surpassing previous methods. In the context of AF2 winning the CASP14 protein structure prediction competition and its developers winning the Nobel Prize, many headlines have claimed that AF2 has solved the PFP. I propose to adopt a more nuanced view of AF2, taking into account the various objectives associated with the PFP and the objects of understanding its solution involves. I will argue that, despite its empirical success, AF2's complexity and opacity limit its capacity to contribute directly to the scientific explanation of the PFP and, consequently, to its scientific understanding. I will present four conditions for scientific understanding mediated by a method: information integration, abilities, the generation of potential explanations and the provision of actual explanations. Based on those, I will show in a review of scientific articles that, despite its opacity, the use of AF2 can, and already has, enhanced objectual and ultimately explanatory understanding of certain research questions in protein biology, even if its inner workings remain mysterious. To make this claim, the interplay between scientists and the predictions plays a significant role, which indicates a new dynamic in scientific understanding, in which explanatory understanding is gained in a two-step adaptive process.

Keywords Machine learning · Scientific understanding · Computational biology · Artificial intelligence

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1 Introduction

The protein folding problem (PFP) has puzzled researchers in the life sciences for the past 50 years. Its solution would revolutionize medicine, considering that proteins figure prominently in the functioning of the body. They fulfill a variety of tasks such as catalyzing reactions, transport, stabilization and signal transmission, and their function is in most cases dictated by their folded structure. Starting from an amino acid sequence, they follow a complex folding pattern. Just how exactly the amino acid sequence and the present physical forces determine their functional, final state and allow for this state to be achieved very quickly, which requires an effective navigation of the vast space of possible intermediary conformations, is still unclear, which is the core of the PFP.

With a lot of experimental effort, the 3D structures of about 200,000 out of hundreds of millions of proteins known by sequence have been determined experimentally¹. This figure has drastically changed since DeepMind introduced its deep neural network (DNN) AlphaFold2 (AF2), a DNN built on the transformer architecture that analyzes evolutionary patterns in amino acid sequences. It predicts spatial relationships between amino acids to assemble highly accurate 3D protein structures. AF2 surpassed all other protein structure prediction algorithms in accuracy and has provided access to 215 million structures in the AlphaFold Database (Durairaj et al., 2023), which led to headlines such as “DeepMind’s protein-folding AI has solved a 50-year-old grand challenge of biology” (Heaven, 2020). Demis Hassabis and John Jumper received the Nobel Prize 2024 for its development. The headline of the corresponding press release reads “They cracked the code for proteins’ amazing structures” and further on “Demis Hassabis and John Jumper have developed an AI model to solve a 50-year-old problem: predicting proteins’ complex structures.”²

However, DNNs are black boxes due to their high dimensionality and parameter-instrumentality (Boge, 2022; Sogaard, 2023). Even if the input and algorithms are known, it is generally not possible to determine which of the high-dimensional input features are used and how they are used, and thus how exactly their highly accurate predictions are made. AF2’s accuracy and the number of predictions available in the AlphaFold Database are thus impressive, but their status in contributing to understanding the underlying problem is unclear. This is also shown in articles such as “Has AlphaFold actually solved biology’s protein-folding problem?” (Hesman Saey, 2022), in which the scientist Julie D. Forman-Kay cautions against the blind use of the predictions:

Erroneous predictions could leave some scientists thinking they understand how a protein works when really, they don’t. Painstaking experiments remain crucial to understanding how proteins fold, Forman-Kay says. “There’s this sense now that people don’t have to do experimental structure determination, which is not true.”

¹ <https://www.rcsb.org/stats/summary>, accessed 15th March 2024.

² <https://www.nobelprize.org/prizes/chemistry/2024/press-release/>, accessed 21st March 2025.

This article is concerned with the question of how far scientific understanding can be gained with the use of DNNs as a research method. AF2 is used as a case study to investigate how scientists deal with the opacity of DNNs and to shed light on the dynamics at play when scientific understanding is gained from their use. Promoting scientific understanding can be considered a core aim of science (De Regt, 2017; Elgin, 2017; Potochnik, 2017). It seems to be difficult for highly accurate, but inexplicable predictions like those we gain from AF2 to directly contribute to the understanding of a phenomenon, in particular if understanding is interpreted as knowing an explanation (e.g. Khalifa, 2017; Strevens, 2013; De Regt, 2017). However, I will show that if we build on conditions for understanding that embed the surrounding research process, DNNs can be said to contribute to the scientific understanding of questions in the research area of protein folding in interesting ways. The kind of understanding I have in mind is objectual in nature (Elgin, 2017; Dellsén, 2020); that is, understanding of a field, subject matter or phenomenon by means of grasping connections, acquiring abilities or enriching the body of knowledge surrounding them. This, I will argue, paves the way to an explanatory understanding of some phenomena surrounding protein folding, even though AF2 cannot provide explanations of the folding mechanism and ultimately solve the PFP. I thus caution against the above-mentioned statements on the solution of the PFP as too optimistic and recommend a more nuanced view on what DNNs can and cannot do. This critically requires a deeper examination of the PFP and of the possible objects of scientific understanding with AF2.

The article proceeds as follows: In Sect. 2, I develop a suitable concept of and conditions for scientific understanding with a method based on a summary of the debate on scientific understanding in general. In Sect. 3, I then discuss the role of these conditions in deep learning specifically. Section 4 introduces the PFP and AF2 as an exemplary phenomenon and the DNN used to investigate it. Finally, in Sect. 5, I analyze the influence that its predictions have had on both objectual and explanatory understanding in the domain of protein folding.

This case study shows that there is a new, interesting pathway to understanding with DNNs that does not require a deep understanding of them, but rather their sensible embedding in the research process, as can be witnessed in protein biology. At the same time, it also shows that a certain amount of caution is necessary when marketing the impressive results of DNNs, as DNN predictions themselves, without further input from highly knowledgeable scientists who develop the DNNs and interpret their results, cannot be said to advance scientific understanding of phenomena, be it objectual or explanatory.

2 Conditions for scientific understanding

Two kinds of understanding are relevant to the following discussion: explanatory and objectual understanding. Explanatory understanding is often equated with understanding-why and understanding-how (e.g., Khalifa, 2017; Strevens, 2013) and its object is a phenomenon or explanans, as in “A scientist S understands why the climate is changing.” In what Strevens calls the simple view, S is said to have “sci-

entific understanding of a phenomenon just in case they grasp a correct scientific explanation of that phenomenon” (Strevens, 2013, p. 1). However, what counts as a *scientific explanation* is debated and there are different accounts, like the deductive-nomological model, causal-mechanistic, and statistical explanations (see Salmon, 1989 for an overview), which could seem to complicate the evaluation of explanatory understanding. I will follow Bokulich (2018) in her explanatory pluralism: different scientific disciplines might prefer different types of explanation, and, within the same discipline, the same phenomenon can be explained with different types of explanation. What counts for this article is that scientists consider it a scientific explanation, the grasping of which leads to scientific understanding.

Objectual understanding is described as the understanding of a subject matter, scientific discipline or a phenomenon, as in “S understands physics” or “S understands evolution”. Objectual understanding has many different objects and sources. These include not only explanations but also the relationships between facts, integration of different kinds of knowledge, and application of relevant concepts to novel situations.

Dellsén (2020) notes that the term objectual understanding is “slightly misleading since the target phenomenon P need not be a single object; rather, it may be a complex system that is itself most naturally described as being composed of several interacting objects.” (p. 3). Depending on the author, objectual understanding is said to concern scientific disciplines as a whole (S understands quantum physics), subject matters (S understands climate change) and phenomena (S understands rainbows). It is advisable to concisely specify the object of objectual understanding, particularly in contexts in which it is compared to explanatory understanding. This becomes important in my case study, and is addressed in detail in Sect. 4.4. The term phenomenon is rather vague and in the present case study on protein biology, explananda are causal and structural information that can be fit into a causal network of information. The corresponding objectual understanding is reflected in a network of information on protein biology as a subject matter or of specific questions therein, while explanatory understanding of the PFP concerns the question why proteins fold the way they do and has to be answered in terms of causal and structural information (details are in Sect. 4.1).³

What is the relationship between explanatory and objectual understanding? Khalifa argues that objectual understanding can be treated as either “(a) explanatory understanding, or (b) being on the right track to having explanatory understanding” (Khalifa, 2017, p. 97), where “‘being on the right track’ means that one does not know the proper answer to a relevant explanation-seeking question, but one has information that would be useful in acquiring such knowledge.” (ibid., p. 98) He argues against a special status of objectual understanding, since it can be subsumed, without loss, under the category of ‘incomplete’ explanatory understanding. In contrast, Elgin argues that objectual understanding is more fundamental:

I do not deny that the sort of understanding why that figures in explanation is real. I simply doubt that it is either as comprehensive or as fundamental as its adherents think. Objectual understanding, I take it, is the core conception of

³ I thank an anonymous reviewer for encouraging me to highlight this point.

understanding. It affords a basis for distinguishing between understanding a topic and knowing particular truths about it. (Elgin, 2017, p. 32)

Objectual understanding can thus be regarded as ‘weaker’ than explanatory understanding, because it does not require a scientific explanation, but also as ‘stronger’, as it concerns a broader body of information. It is outside the scope of this article to develop a nuanced position between these two extremes. It suffices for my analysis to assume that there are good arguments that both objectual and explanatory understanding are important kinds of scientific understanding and that the possibility to provide either of the two with DNNs would be remarkable. Furthermore, both Khalifa’s and Elgin’s framings of objectual understanding allow for explanatory understanding to result from it – in Khalifa’s case by eventually completing it, if it is only on the right track, and in Elgin’s case ‘particular truths’, i.e., explanations, about a topic follow directly from objectual understanding.

The main sources of scientific understanding mentioned in the literature are explanations (De Regt, 2017; Khalifa, 2017; Strevens, 2013), potential explanations (Bokulich, 2014; Khalifa, 2017; Lipton, 2009), abilities (De Regt, 2017) and new information that is integrated into the body of information on a phenomenon/domain (Dellsén, 2020; Elgin, 2017). I propose to capture these sources in four sufficient conditions under which the results of applying a method *M* contribute to the scientific understanding of a phenomenon *P*. This will allow me to evaluate the scientific understanding that is potentially gained from the use of deep learning in the research process in my case study.

The most obvious and widely accepted way to achieve scientific understanding is by grasping a scientific explanation. A method *M* that directly outputs an explanation of a phenomenon *P* thus contributes to understanding directly, which is captured in the explanation condition:

Explanation Condition (EC): An *M*’s results contribute to the scientific understanding of *P* if they constitute a scientific explanation for *P*.

EC captures the straightforward cases in which explanations directly constitute understanding. For example, the results of the method of statistical data analysis can allow scientists to formulate causal or mechanistic explanations of natural phenomena. The condition also accommodates model-based explanations, where the explanatory power resides in the model’s ability to represent complex relationships, generate predictions, or clarify causal mechanisms, as noted in Lawler (2024): “there are model explanations where the non-propositional model itself is the explanation” (p. 302). An example would be the Plate Tectonics model that describes the Earth’s lithosphere as being divided into several large, moving plates that float on the semi-fluid asthenosphere beneath. These tectonic plates interact at their boundaries, which can explain various geological phenomena like earthquakes or the formation of mountains.

The fulfillment of the EC by a method is rare. Often, methods do not directly output explanations, but results that have to be interpreted by scientists. Many models make idealizing assumptions that have to be justified to arrive at a comprehensive explanation and statistical results need to be carefully analyzed to differentiate

between correlation and causation. A second important source of understanding is potential explanations generated on the basis of the results of a method. I subsume under this debate how-possibly explanations, potential explanations, and answers to what-if questions⁴. Coming up with potential explanations and then testing them is an integral component of scientific practice. Lipton (2009) argues that these explanations contribute to actual understanding, namely knowledge of possibility, because “information about other worlds illuminates the actual world” (p. 50). But there are also potential explanations concerning the actual world that “are about what is possible in this world, not about facts in some other, possible, world” (Grüne-Yanoff, 2013, p. 852). How-possibly explanations are different from hypotheses because “we can give a true explanation of how possible that, after empirical investigation, we reject as the actual explanation.” (Forber, 2012, p. 299), while hypotheses can be disconfirmed by empirical testing.

Khalifa also notes the epistemic value of potential explanations when he states that “considering plausible potential explanations of a phenomenon enhances one’s objectual understanding.” (Khalifa, 2017, p. 102). He also notes that not only considering them, but also “disconfirm[ing] otherwise-plausible potential explanations through sound scientific method and evidence” (Khalifa, 2017, p. 103) contributes to objectual understanding: “answers to what-if questions are consequences of correct explanations, some of which are essential to scientists’ evaluations of which explanations are correct” (Khalifa, 2017, p. 145). While the question of whether they contribute to explanatory understanding has to be answered negatively, because it is concerned with how-actually explanations, they can plausibly be argued to contribute to the objectual understanding concerned with the discipline into which the phenomenon to be explained falls. As a second condition, I therefore propose the potential explanation condition:

Potential Explanation Condition (PEC): An M’s results contribute to the scientific understanding of P if they are used to develop potential explanations for P.

The PEC applies to all methods that allow scientists to develop new potential explanations. They also include new numerical models with assumptions that are not yet confirmed to hold, but that would lead to correct explanations after empirical testing of the assumptions. The PEC also often applies to methods used in the early stages of research, in which many competing explanations are developed and constitute new hypotheses to be followed up on experimentally (cf. Bokulich, 2014). In this sense, methods that fulfill the PEC can contribute valuable objectual understanding that, together with empirical verification, can lead to explanatory understanding.

As a third condition for the results of using a method to contribute to scientific understanding, I propose their potential to provide researchers with new abilities or know-how that enhance their understanding of the phenomenon, i.e., the ability to

⁴There is, of course, an important difference between hypothetical and counterfactual reasoning, as noted below, counterfactual reasoning requires the grasp of mechanisms surrounding a phenomenon, while hypotheses provide possibilities. I thank an anonymous reviewer for highlighting this distinction.

use the method contributes to the understanding of P. This refers to practical abilities as well as to the ability to draw what-if inferences or to infer connections that would not have been possible without the use of the method. Elgin characterizes the abilities related to understanding as knowing “how to draw the inferences and perform the actions that the understanding licenses” (Elgin, 2017, p. 39). De Regt (2017) also emphasizes the importance of practical abilities for understanding. A method that provides scientists with such abilities contributes to scientific understanding.

Ability Condition (AC): M contributes to the scientific understanding of P if its use equips scientists with new abilities or know-how related to the investigation of P.

De Regt (2017) emphasizes for example the importance of visualizability for understanding. Methods that enable scientists to visualize information related to P in a novel way can contribute to the scientific understanding of P by providing scientists with the ability to generate new insights and to draw new inferences. The ability to use a microscope constitutes a new ability for scientists to see and interpret microscopic structures. A model of a disease mechanism can provide doctors with the ability to decide on a treatment.

The last condition accounts for the role new information plays for understanding, even if this information does not directly figure in abilities or (potential) explanations. As scientists collect and analyze data, they enrich the objectual understanding of a phenomenon by either integrating new data into the body of information surrounding the phenomenon or by identifying (causal) relationships between different parts. This enriches the body of scientific knowledge of the phenomenon and forms a basis for new (potential) explanations.

It is important that not any new information can contribute to understanding, as this would open the door for data that actually hinders the research process, like fabricated data or p-hacked data, to be included, even though these techniques lead to less understanding. Coherence is one criterion to ensure meaningful integration that excludes random, unrelated information to count for an increase in the understanding of P. A method that leads to a useful enrichment of the existing body of knowledge of P fulfills the information integration condition (IIC):

Information Integration Condition (IIC): An M’s results contribute to the scientific understanding of P, if they are used to gain new information on P that can be meaningfully integrated into the existing body of knowledge of P.

The IIC also subsumes how-possibly explanations in the sense of Forber’s global how-possibly explanations, which I would call purely hypothetical explanations, that is, explanations that are not at all empirically supported: “we can successfully explain how possible without any empirical support; we need only show that some outcome or pattern is consistent with a specific set of information.” (Forber, 2012, p. 39) In this sense, they enrich the objectual understanding of a domain without leading to predictions. The IIC is supposed to capture what Kelp means when he states that “non-explanatory knowledge can improve understanding” (Kelp, 2015, p. 3807). Kelp

calls “the union of a phenomenon’s description and story its ‘full account’.” (Kelp, 2015, p. 3809), including all propositions that are true descriptions of P as well as the set of true propositions about P’s “place in a broader nexus of phenomena” (ibid.). While it is probably reasonable to soften the requirement of truth to ‘true enough’ propositions in the sense of Elgin (2017), it captures the contribution that integrating new information makes to the scientific understanding of phenomena.

The role each of these conditions plays in the scientific process is visualized in Fig. 1. It is important to note that this is an idealized scientific process. In practice, the scientific process is rarely linear and highly iterative, with data, abilities, explanations and hypotheses playing important roles at various stages. The conditions are highlighted in specific contexts to emphasize their primary contributions to scientific understanding. Dashed gray arrows indicate exemplary relations between the conditions and an exemplary scientific action that can be performed to fulfill the other condition. A successful application of a method (AC) can, for example, lead to new data analyses, fulfilling IIC, and new explanations, fulfilling EC. Explanations, in turn, can lead to new practical abilities, as can the know-how development from methods fulfilling IIC, leading to both of them potentially fulfilling AC. Methods fulfilling the IIC can, via hypothesis development based on the observations or new data, fulfill the PEC. While I treat the conditions separately to highlight their specific contributions to understanding, they are not mutually exclusive and, in practice, one method can fulfill multiple conditions simultaneously. It is, for example, possible that one and the same method fulfills both data and hypothesis generation at the same time, which will be seen in the case study as well.⁵

To apply these conditions to a practical example, consider a situation in which climate scientists research coral health. They would typically begin with explora-

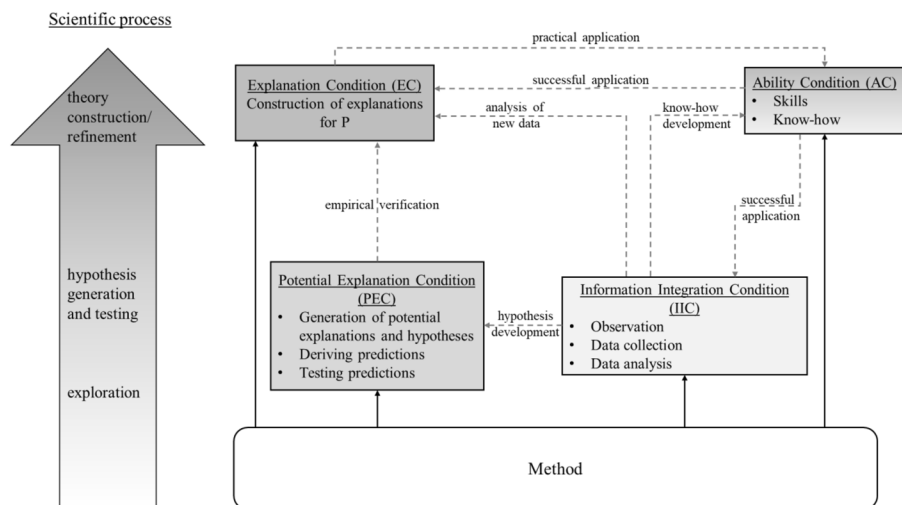


Fig. 1 The developed four condition for understanding with a method in an idealized scientific process. Dashed, gray arrows indicate exemplary relations between the condition via scientific actions

⁵ I thank an anonymous reviewer for suggesting that I make this point clearer.

tion, using methods to gather data on sea temperature, acidity and coral reef health across different locations. Techniques such as remote sensing, satellite imagery, field surveys, and in-situ measurements could be employed to collect the relevant information. Additionally, statistical analysis and data mining methods can be used to analyze patterns and correlations between coral health and these environmental factors. These methods fulfill the IIC by providing a comprehensive body of information that helps scientists detect and integrate patterns related to coral health. The PEC would be fulfilled by methods that allow them, based on these patterns, to develop specific potential explanations and experiments designed to test them, in our example this could be the hypothesis that higher ocean acidity leads to slower growth in corals, which could for example be gained from a statistical method. This could be tested with the method of laboratory experiments gathering specific data on corals grown in aquariums with different pH levels. Another way could be the use of predictive modelling that simulates different scenarios like different levels of sea temperature or CO₂ concentrations. EC requires a method that develops explanations, usually based on the objectual understanding gained in the previous stages. In our example, an explanation could be that higher acidity causes a reduction of the carbonate ion concentration, which is required for coral skeleton formation, and therefore leads to slower growth of the corals. While fulfillment of the EC is one of the ultimate goals of science, the process towards EC often crucially involves both IIC and PEC and, thus, objectual understanding. The AC focuses on the abilities, know-how, and intelligibility scientists gain from applying and interacting with methods. In the context of coral health research, AC could involve the practical expertise required to operate specialized instruments like remote sensing equipment, design and execute laboratory experiments, or to build and interpret predictive models. Additionally, methods that fulfill AC contribute to intelligibility by helping scientists understand the relations between environmental factors and coral health, which could equip them with the know-how required for identifying potential interventions or policies for mitigating the negative effects of ocean acidity.

In what follows, I will analyze how the developed conditions fare in light of understanding with DNNs, or black boxes. While DNNs can process large volumes of data and could potentially uncover complex patterns, they face challenges in fulfilling the outlined conditions. Regarding IIC, DNNs may provide data-driven insights and correlations, but their black box nature can hinder their integration into a body of information, as the models often lack transparency in how specific factors influence the predicted outcome. As for PEC: DNNs may generate predictive potential explanations based on patterns, but without a clear interpretation, they can remain difficult to test empirically. Stronger versions of the PEC, for example enabling counterfactual reasoning, do not seem to be straightforwardly possible with DNNs, as the mechanisms stay opaque. EC is particularly challenged by DNNs, as these methods often provide predictions without offering clear, causal explanations. And, while DNNs can contribute to the AC by providing researchers with the ability to automate data analysis and pattern recognition, what set of abilities exactly their use provides is difficult to assess due to their black box properties as well.

3 Understanding with black boxes

DNNs have in recent years emerged as powerful tools that allow scientists to identify complex patterns in, and make predictions from, vast datasets. Their impressive performance led to their application in most scientific disciplines. They are black boxes due to their high dimensionality and the instrumentality of their parameters. It is not only a challenge to understand them but also to understand a phenomenon based on their predictions because of their opacity. Humphreys (2009) defines a system as “essentially epistemically opaque for [a cognitive agent] X if and only if it is impossible, given the nature of X, for X to know all of the epistemically relevant elements of the process” (p. 618). Boge (2022) shows that two dimensions of opacity can be distinguished for DNNs: it is not only opaque how they learn (h-opacity), but also what they learn (w-opacity), which is “something that connects more closely to questions of understanding ‘the world’ rather than ‘the machine.’” (p. 63) W-opacity seems to influence the ability of scientists to understand something *with* a DNN – if it is unclear on what patterns and relationships in the data the predictions are based, there seems to be no basis for understanding, neither of the predictions nor of the researched phenomenon.

There are two possibilities for interpreting M and P from the conditions for understanding developed in Sect. 2. First, the DNN can be seen as the phenomenon that is supposed to be explained. This is not the subject of this article, but part of the line of research in eXplainable AI (XAI). Here, XAI would be the method M employed to understand phenomenon P, the DNN. What I want to discuss in this article is what happens when we select the DNN as M in order to further the scientific understanding of P, which refers to protein folding in my case study. This is not to say that research in XAI is no valuable path to scientific understanding. However, in particular in the case of AF2, XAI methods have not been used to better understand the DNN. Instead, AF2 was used directly to research phenomena and thus as a method, not the phenomenon.

The question I am concerned with is thus in how far a black box method’s predictions further or limit scientific understanding. In this section, I will first discuss some philosophical contributions to this subject (3.1) and then present possible pathways to understanding with DNNs (3.2).

3.1 Setting the stage

In what follows, I focus on the question of whether understanding something with a poorly understood method is possible. It is obvious that an answer to this question strongly depends on the concept of understanding one adopts. As argued in the previous section, scientific understanding requires at least some contribution to the process of scientific investigation, be it via directly explaining (EC), or via constructing potential explanations (PEC), providing information that is in some way valuable for the first two (IIC), or equipping scientists with abilities (AC).

Sullivan (2022) begins her discussion on the possibility of understanding phenomena with machine learning models by stating that her “arguments do not so much trade on any positive notion of what understanding or explanation is, but on whether, given a lack of information, it is still possible to gain insight about a phenomenon” (p. 111). She takes

understanding with DNNs to be possible to the extent that they are not associated with link uncertainty, “that is, a lack of scientific and empirical evidence supporting the link that connects the model to the target phenomenon” (p. 112). She argues that a DNN that predicts melanoma based on visual data is, for example, so clearly related to the phenomenon and general medical practice that its black box components (e.g., exact weights) are not hindering the possibility to understand what the classification is based on, namely on visually inspecting suspect skin changes.

On the other hand, she shows how a DNN that predicts sexual orientation from photographs is not connected to anything known to apply in reality. How the data it uses result in strikingly correct predictions is not only highly black boxed, but additionally subject to link uncertainty. Sullivan, thus, makes the possibility of understanding with DNNs dependent on the information that surrounds them, on how they are embedded in the research process. If their embedding allows a clear link between the model and the researched phenomenon, understanding the phenomenon better is possible even if the implementation of the model is not well-understood.

In the melanoma case, there is a clear link between the model’s method and the standard physician’s method, while only the exact implementation of the method is black boxed:

Implementation black boxes do not get in the way of understanding phenomena in the melanoma case because the model is operating within a background of existing scientific understanding. So, although we do not understand all the low level details about how the model works, and even though the model is complex in its data-points, we gain understanding of skin classification nonetheless. (p. 126)

In a strong reading, the last sentence suggests that the EC is directly fulfilled by the DNN. I would argue that the IIC and PEC are fulfilled in this example. The model does not provide understanding of skin classification as direct explanations, which would be required to fulfill EC. Instead, it delivers a high volume of accurate melanoma classifications (IIC) and enables the derivation of hypotheses from correlations in the data (PEC). To develop an explanation, more than the method’s predictions are necessary – they would have to be analyzed by trained professionals who might then be able to identify formerly unknown patterns in the data based on the predictions. Sullivan seems to agree with that weaker view when she writes that the model can “help physicians gain understanding about why certain medical interventions are relevant” (ibid), which would fulfill the IIC, as new information is added to the body of information related to melanoma classification and could, arguably, also fulfill AC, if the scientists can use the predictions to gain new know-how on identifying melanoma cases. She continues stating that the predictions bring scientists in a position in which they “can discover new visual patterns that are highly correlated with health or disease” (ibid), which would fulfill PEC, as the DNN’s predictions would directly enable scientists to construct new potential explanations: “This can further understanding, especially once these newly discovered patterns undergo further investigation.” (ibid). In Sullivan (2023) she also emphasizes “that it is possible that these features are indicators of melanoma” (p. 7), making clear that real scientific progress

can be made with this DNN. Sullivan's account is thus compatible with DNNs as method being potentially able to fulfill IIC, PEC, AC and, in a strong reading, potentially EC, on the condition that their use is sufficiently linked to the phenomena under investigation.

Răz and Beisbart (2022) note that Sullivan does not specify which notion of understanding she argues for, but interpret her to argue for explanatory understanding: "she assumes a close connection between understanding and explanation: '[i]n a slogan: explaining why helps us to understand why' (p. 110). We thus take it that her main interest is in explanatory understanding" (p. 3). They argue explicitly "contra Sullivan" that only a – as they see it – "weaker" form (p. 16), objectual understanding, can be gained from DNNs as long as they are not better understood, as deriving actual explanations from DNN predictions would require a much deeper insight into their inner workings. Their main argument is that the functions that DNNs compute are too complex for us to understand and therefore unknown, which can lead to problems with instances that are misclassified by the DNN, as is often seen with out-of-distribution samples. This leads them to defend a more generalized skepticism towards understanding with black boxes.

Răz and Beisbart clearly argue against EC being possibly fulfilled with DNNs as a method, as they make clear from the beginning when they specify that "predictive success is not sufficient for explanatory understanding unless a predictively accurate tool provides at least some explanatory information." (p. 10) They do, however, support the fulfillment of IIC when they remark: "ML models can lead to some objectual understanding, e.g. by establishing correlations, or by simply adding to knowledge of a domain of things." (p. 12) Compared like this, the points made in both articles are not too far apart: they both acknowledge the role of DNNs in enhancing certain aspects of understanding by possibly fulfilling IIC and PEC, and differ on the stances they take towards the EC, which Sullivan supports under very specific conditions while Răz and Beisbart outright reject it. The fact that methods can also equip scientists with know-how, fulfilling the AC, is only briefly touched upon by Sullivan and not mentioned by Răz and Beisbart.

3.2 Pathways to scientific understanding with black boxes

Sullivan and Răz and Beisbart seem to agree that DNNs can fulfill both the PEC and IIC – their predictions can serve as new hypotheses for future experiments and they can provide information that deepens the objectual understanding of the domain surrounding the phenomenon in question. Below, I will argue that the fulfillment of AC is also an important contribution of DNNs to scientific understanding.

However, scientists arguably look for explanations of phenomena. It would thus be interesting to see which paths to fulfill EC with DNNs are possible. The direct path to achieving EC is challenging; as discussed above, it is a hallmark of DNNs that they are opaque with regard to how they arrive at predictions. As noted in the example of melanoma prediction, the generation of explanatory understanding appears to involve researchers actively following up on the hypotheses and information generated by the DNN with new experiments.

It should be remarked here that one should be careful not to count every prediction any DNN makes as contributing to IIC or even PEC. Both authors emphasize that the black box has epistemic consequences and while Sullivan is more optimistic about the potential of understanding with black boxes, she demonstrates that clearly not any prediction leads to a rise of understanding, as in the example of the unscientific DNN that predicts sexual orientation. Her condition of minimizing link uncertainty is one plausible proposal to ensure the potential of understanding real-world phenomena with DNNs and additional conditions might be necessary. For instance, it is crucial to reliably exclude cases where DNNs pick up on spurious correlations that lead to *Clever Hans* behavior – that is, when they make correct predictions for the wrong reasons (Boge, 2024).

While I sideline the topic of XAI in my discussion, this is not to say that XAI is not important with regard to this question. Many philosophers have argued for the need for interpretability (e.g., Beisbart & Räs, 2022; Pérez, 2019, 2024; Zednik & Boelsen, 2022). Understanding DNNs themselves better certainly is a worthwhile endeavor that can enhance the understanding of the researched phenomenon as well. Some have also argued that data-driven methods can support causal inferences, for example by employing strategies to identify difference-makers (Pietsch, 2016) or with techniques that extract structural equation models (Buijsman, 2023). What I am concerned with is the way DNNs are directly used by researchers investigating phenomena with them, and, thus, the question how scientific understanding can be gained *with* or even *despite* opacity.⁶

In the next two sections, I will introduce the protein folding problem and AlphaFold2 as a case study of a scientific process with extensive usage of a DNN. This will show that the distinction between objectual and explanatory understanding is often fruitfully applied to different steps in the scientific process and that a valuable contribution to objectual understanding in the form of fulfilling PEC, IIC and AC can be just the missing piece to explanatory understanding and fulfilling EC. It will show that the objectual understanding they provide should not be disregarded as ‘weak’, but rather is a crucial part of the research process with DNNs.

4 The protein folding problem and AlphaFold2

Proteins are fundamental to life: they play crucial roles in various bodily functions, including enzymatic activity, immune response, signal transmission and structural support. Their 3D structure oftentimes indicates their function and is therefore very important information for scientists. While the amino acid sequences of most proteins are known thanks to substantive progress in genomic sequencing, their folded structure can only be determined in lengthy and complicated experiments making use of X-ray crystallography, nuclear magnetic resonance (NMR) spectroscopy or cryo-electron microscopy (cryo-EM), as a general folding mechanism that would predict their folded structure has not yet been found. This is the challenge formulated in the protein folding problem.

⁶I thank an anonymous reviewer for pushing me to make this point clearer.

In Sect. 4.1, I will review the protein folding problem, to elucidate the context of the development and success of AF2 in Sect. 4.2. In Sect. 4.3, I will discuss some of the DNN's limitations and in 4.4 the relationship between the PFP and AF2.

4.1 The protein folding problem

The protein folding problem (PFP) has puzzled researchers since the 1950s, when, in a series of experiments, Christian Anfinsen found that all information that is required to fold a protein is encoded in its amino acid sequence, a principle known as Anfinsen's dogma (Gomes & Faisca, 2019, p. 13). Levinthal's paradox highlights the apparent contradiction between the vast number of possible conformations a protein could theoretically explore and the fact that proteins fold rapidly into their native state, suggesting that folding follows specific pathways rather than randomly exploring different conformations⁷. As proteins come in many different sizes and can form very complex and unique folded structures, a solution to how exactly all this can be encoded in the amino acid sequence is challenging and remains unsolved to this day. The PFP is an interdisciplinary problem considering methods and perspectives from physics, chemistry and computational biology.

Commonly, the PFP is split into 3 subquestions (Dill et al., 2008, 290; Gomes & Faisca, 2019, 2):

Thermodynamic PFP (tPFP): Which forces determine the folded structure?

Kinetic PFP (kPFP): How is the folded structure achieved so quickly?

Computational PFP (cPFP): How can the folded structure be determined from the amino acid sequence?

The tPFP centers on how the amino acid sequence adopts its 3D structure by seeking the global minimum of its Gibbs free energy landscape. It can be formulated as the question 'Why are proteins folding the way they do?' An answer to this question needs to account for various molecular interactions between the amino acids, including hydrogen bonding, hydrophobic packing, as well as electrostatic and van der Waals interactions, which are known to be required to achieve a stable and energetically favorable conformation. That only the amino acid sequence and the physiological conditions determine a protein's structure is called Anfinsen's dogma, or the thermodynamic hypothesis:

the three-dimensional structure of a native protein in its normal physiological milieu (solvent, pH, ionic strength, presence of other components such as metal ions or prosthetic groups, temperature and other) is the one in which the Gibbs free energy of the whole system is lowest; that is, the native conformation is determined by the totality of the interatomic interactions and hence by

⁷ Levinthal (1969) shows in a thought experiment that even a small protein consisting of 150 amino acids would have 10^{300} possible conformations if the bond angles were known to the tenth of a radian (p.22).

the amino acid sequence, in a given environment (Anfinsen 1973, quoted from Gomes & Faísca, 2019, p. 13)

Despite the known factors involved in addressing the why-question, the vast conformational space and complex energy landscape – characterized by numerous local minima – complicate efforts to provide a general answer.

The kPFP centers on the kinetics of protein folding, posing the question ‘How do proteins fold so quickly?’ It seeks to enlighten the pathways and intermediate states that proteins go through during the folding process, asking how proteins achieve their functional conformation so rapidly despite the multitude of intermediate conformations. Together, the tPFP and kPFP focus on the folding *mechanisms*, examining both the driving forces behind the final structure and the dynamic processes that allow rapid folding, while the cPFP focuses on predicting the outcome of this mechanism.

An important property to note is that there are many proteins that have no well-defined tertiary structure (intrinsically disordered proteins, IDPs) or parts of proteins that lack such structure (intrinsically disordered regions, IDRs). This can be important, for example, for “proteins involved in signaling processes that are able to accommodate a considerable ‘fuzziness’ within their structure” (Gomes & Faísca, 2019, p. 6), where the lack of structure is advantageous for binding to multiple targets. In the case of IDPs, “[a]s a trade-off for the lack of a unique three-dimensional structure, IDPs generally sample a variety of conformations that are in dynamic equilibrium under physiological conditions” (ibid, p. 7).

There are also proteins that have multiple conformations depending on factors such as the concentration of certain molecules in the cell or whether they are bound to a ligand. The cPFP faces the challenge of predicting dynamic structural ensembles for IDPs, as their conformations may change depending on specific environmental conditions or interactions with other molecules, requiring additional information to determine which conformation is favored in a given context. Of course, the existence of multiple conformations also complicates answers to the tPFP and kPFP. The tPFP is complicated because IDPs often do not have a well-defined native state under physiological conditions. Instead of a single global energy minimum, IDPs exhibit an energy landscape that is characterized by an ensemble of interchangeable conformations. This suggests that the thermodynamic principles governing structured proteins may need to be adapted to account for their flexibility. The solution to the kPFP is challenged because IDPs do not fold into a single, stable conformation but instead explore multiple dynamic states. Their folding pathways could be influenced by environmental cues or interactions with other molecules, as is seen for example in binding-induced folding, which introduces additional layers of complexity to the kinetic landscape.

The PFP, in its entirety, demands an explanation of *why* and *how* proteins fold the way they do. This entails more than solving a single computational task. Rather, it requires an integration of thermodynamic principles, kinetic models, and computational techniques capable of capturing the structural and functional diversity of proteins under physiological conditions. The tripartite structure of the PFP, encompassing the tPFP, kPFP, and cPFP, reflects the multifaceted nature of this challenge. It is not enough to show that a given protein structure can be predicted accurately

(cPFP), nor to enumerate the molecular interactions stabilizing a folded state (tPFP), nor even to trace the steps through which folding occurs (kPFP). A satisfying resolution of the PFP must bring these dimensions into coherence, ideally offering a unified account that explains how the sequence, structure, dynamics, and function of proteins relate across contexts. The complications arising from IDPs show that there are no single native states for every protein and that the solution to the PFP has to take the cellular biochemical context into account. For the cPFP, this means that a correct prediction is not necessarily a single conformation, but can be multiple conformations associated with conditions under which they most likely occur.⁸

The PFP is thus not solved solely by amassing objectual understanding of folded structures, although, as Elgin (2017) argues, explanations often arise from objectual understanding. Scientists become able to explain phenomena when they grasp a sufficiently broad and systematically organized body of information.

4.2 AlphaFold2

The lack of a general solution to the PFP motivated the development of AF2 and similar ML models. AF2 is a transformer network that combines several new technologies to arrive at very accurate protein structure predictions from amino acid sequences, evolutionary information and templates of already known structures. The trained model can predict new structures rapidly and accurately. In comparison, it is very time-consuming to determine protein structures experimentally and it requires many preliminary assumptions and huge amounts of computing power to do so with physically (more) accurate computer simulations.

Details of how AF2 works are in Jumper et al. (2021) and the corresponding appendix. A schematic overview is presented in Fig. 2. AF2 begins with an amino acid sequence as input from which a Multiple Sequence Alignment (MSA) is generated, which identifies conserved regions by aligning the sequence with similar sequences from various organisms. Additionally, where available, templates of known protein structures are incorporated to enhance the pair representations, providing structural context based on homologous proteins.

The model starts with two distinct representations: the MSA representation, which captures information about individual amino acids, and the pair representation, which is derived from the MSA and captures relationships between pairs of amino acids. The Evoformer processes these representations using attention mechanisms and information recycling to refine the data for enhanced accuracy. After processing, the model produces both single and pair representations. These refined representations

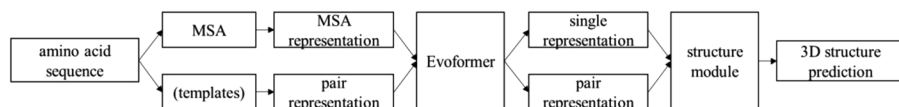


Fig. 2 Schematic components of AlphaFold2

⁸ Again, thanks are due to an anonymous reviewer who emphasized the importance of being clear on the form a solution of the PFP would take.

are then fed into the structure module, composed of eight blocks, which predicts the protein's 3D structure by mapping the data to spatial coordinates. In the last step, the technique AMBER (Assisted Model Building with Energy Refinement) is used to correct atomic positions based on physical interactions between them and taking into account molecular mechanical principles of energy minimization. The final output is five predicted protein structures for each input sequence. Each predicted structure is accompanied by a confidence score, known as the predicted Local Distance Difference Test (pLDDT), which assesses the reliability of the structural prediction. The output also includes the predicted Root Mean Square Deviation (RMSD) values, which indicate the structural variation among the predicted conformations.

AF2 significantly outperformed other structure predictors in the CASP14 competition, achieving a median Global Distance Test score of 92.4, which approaches the accuracy of experimental methods. Furthermore, the AlphaFold Database contains structures for most known human proteins and many more. It raised the number of proteins for which structure data are available from 200,000 to 200 million. AF2 was readily adopted by the research community: the number of journal articles and preprints surged as soon as the detailed paper with source code was published in July 2021 from less than 25 to more than 50 preprints and just as many journal articles in March 2022⁹. Already in August 2021, ColabFold (Mirdita et al., 2022) was developed, which allows users to predict protein structures with AF2 without requiring extensive computational resources. In December 2021, AlphaFold Multimer followed, which extends AF2 to the prediction of protein complexes, allowing a more in-depth analysis of more complex protein structures and protein-protein interactions.

4.3 Limitations of AlphaFold2

While its successes are impressive, it is difficult to explain how AF2 achieved them. Its training did not include any explicit general principles of biology, chemistry, or physics, such as specific bond angles that have to be respected, the hydrophobic properties of certain residues, and hydrogen bonding. Indirectly, AF2 has some principles encoded. AF2's attention mechanism allows it to learn relationships that indirectly reflect physical constraints (e.g., bond angles, distances between residues), even if these are not hardcoded. It has a biological basis, as its predictions are based on evolutionary information in the form of multiple sequence alignments (MSAs). Its optimization process is said to indirectly take into account a minimization of the free energy of the structure, which is a principle from chemistry and physics (Jumper et al., 2021). Of course, AF2 was trained on many folded structures from the Protein Data Bank (PDB) that follow these principles. It is possible that AF2 generalized principles from these structures, but its heavy reliance on existing structures has also been criticized (Wang et al., 2024, p. 5). Roney (2022) presents arguments that AF2 did indeed learn a biophysical potential function that determines the folding but relies on MSAs to find a "good initial guess" (p. 51) on which it can base the prediction, basically orienting itself in the determined potential, which considerably reduces the time required to compute the structure prediction.

⁹Data from the figure "AlphaFold Mania" in Callaway (2020), p. 235.

One limitation of AF2 arises from its reliance on MSA, that is, evolutionary relations. This information is a good structure indicator for a simple reason: evolutionarily related proteins generally have a similar structure. Thus, if the structure of one protein from an evolutionary family is known, it can provide insights into the structures of other proteins in that family. A problem arises for proteins that are not closely related to others, like orphan proteins, mutated proteins or proteins that evolve rapidly, like antibodies, because there is little evolutionary information available, leading to a small MSA. Jumper et al. report that AF2's accuracy is reduced considerably with small MSAs. Moreover, the generation of MSAs can be very time consuming, which makes them a bottleneck for large-scale analyses (Fang et al., 2023). While newer implementations like ColabFold use MMseqs2 to speed up this process, the computational and time demands for generating deep MSAs remain significant, especially when analyzing large datasets or complex proteins.

There is also a training bias, as the PDB contains predominantly well-known proteins, such as globular enzymes and structural proteins, and they are often characterized under specific experimental conditions. This could influence AF2's ability to generalize to less common protein types, such as intrinsically disordered proteins (IDPs) or membrane proteins, which are underrepresented in the database. Furthermore, some structures in the PDB are derived under non-physiological conditions, as proteins need to be crystallized in preparation for experimental methods, which means that they do not necessarily reflect the native state. As a result, AF2 predictions could favor conformations that are observed in experimental conditions instead of *in vivo*.

Another limitation is the static nature of AF2 predictions. Researchers are interested in the folding pathway, as reviewed above in connection with the PFP. What AF2 does is “analogous to interpreting a movie by fast-forwarding to the final scene without first watching the previous two hours; we know how it ends, but we don't know why” (Chen et al., 2023, p. 2). Closely related to the static predictions is the limitation that AF2 might predict only one of several conformations. As explained above, many proteins have multiple conformations, for example depending on whether a ligand is present:

AlphaFold2 was trained on PDB, this specializes the network in predicting structures as they would be found in the PDB, which may not be the natural fold state of that protein. Some of the PDB folds only happen in known special conditions, such as the presence of another protein, these special conditions are usually documented during the solving of the fold. In such a case AlphaFold2 will predict the folding as it is found in the PDB, but is unable to provide the special conditions in which the particular fold happens. (Marcu et al., 2022, p. 5)

The dynamic aspects of protein structure are important factors for understanding protein function and interactions, as also noted in Agarwal and McShan (2024): “AlphaFold performs well only in detecting a single structural snapshot (the ‘ground state’ structure)” (p. 956). While AF2 provides highly accurate single predictions, it does not account for the full range of conformational flexibility that is present in living organisms. This limits its utility for studying proteins whose functional mechanisms

rely heavily on their ability to adopt multiple conformations. In a study of apo-holo pairs¹⁰ of conformers, it was found that AF2 was not able to reproduce the structural diversity associated with these proteins. This limitation can however also possibly provide insights, “main-chain flexibility associated with apo-holo pairs of conformers negatively correlates with the predicted local model quality score pLDDT, indicating that pLDDT values [...] could be used to infer local conformational changes linked to ligand binding transitions.” (Saldaño et al., 2022, p. 2742). The inability to account for proteins with multiple conformations has implications for the practical use of AF2 predictions. For example, in drug discovery, understanding the conformational flexibility of a target protein is essential for designing inhibitors that bind only with specific conformations. Similarly, AF2’s inability to model folding pathways limits its utility in studying protein misfolding diseases, where intermediates play a pathogenic role.

Agarwal and McShan (2024) discuss four ways in which AF2 predictions can be misleading: “(1) inaccurate secondary, tertiary and/or quaternary structure in regions where AF2 predicts low to moderate confidence, (2) inaccurate structure in regions where AF2 predicts high model confidence, (3) correct backbone structure but incorrect fine details (that is, side chain rotamer placements), and (4) correct backbone structure for individual domains but inaccurate placement of domains relative to each other.” (p. 952) They note that while (1) and (4) are indicated by the two error estimates provided by AF2, (2) and (3) can occur without any indication. This is, of course, a problem, and highlights the need for additional experimental confirmation of AF2 predictions.

AF2’s ability to produce highly accurate static predictions without explicitly encoding biophysical principles could suggest that understanding can, at least partially, emerge from predictive accuracy and the capacity to generalize patterns from data. However, its reliance on MSAs and its limitations in modeling dynamic aspects seem to limit the scope and depth of this understanding considerably. It predicts accurate folded states (most of the time), but the folding pathways and mechanisms remain opaque. This static view of protein folding highlights a gap that seems to be necessary to overcome for a deeper grasp of the roles proteins play in biology. Moreover, its reliance on training data from the PDB, with its biases and limitations, suggests that understanding with AF2 is shaped by empirical constraints and has limited generality.

4.4 AlphaFold2 and the protein folding problem

While AF2 is a significant advancement in computational biology, it should be obvious from the previous discussion that it is not even conceptually possible for AF2 to solve all three aspects of the PFP. As it does not output folding mechanisms, but static folded structures, it is not a candidate for providing such a general solution (e.g., Chen et al., 2023), as the tPFP and kPFP specifically ask for mechanisms. AF2 addresses the subquestion cPFP, for which it might seem to provide an answer along the lines of ‘Folded structures can be computed by AF2 from amino acid sequences

¹⁰ Proteins with and without a bound ligand, respectively.

with the help of MSA information and a specific transformer architecture.’ This explanation is, however, not convincing, as it directly invites the next question: ‘*how* does AF2 compute the predictions?’ – a detailed answer to which is not available due to the DNN’s opacity. Not only is this explanation not convincing, it is not even true, as AF2 predictions have the limitations discussed above, such as the inability to handle all proteins effectively and uncertainty that is not always reflected in the provided error measures. Even if one were to accept its computational accuracy, the highly complex architecture of AF2 prevents clear inferences about the underlying general computational processes, i.e., critical information is missing, which would be required to fully understand the cFPF.

The explanatory scientific understanding of the protein folding problem is thus not directly advanced by AF2. But what about more modest goals? Could AF2 be of any help in understanding how certain proteins fold? How they function? Can it give insights into the protein universe? The answer is clearly yes. As already mentioned, AF2 led to the publication of many articles that demonstrate its diverse use cases across various research fields.

I have already mentioned in Sect. 2 that the object of objectual understanding has to be clearly specified in order to gain clear insights into what was understood. In the case of AF2, the object is often taken to be protein folding as addressed in the PFP, as the quotes in the introduction demonstrate. However, according to my analysis, the PFP is not the right goal of understanding with AF2. But several more concrete objects have been researched in the context of AF2 and, in what follows, I will demonstrate that the application of AF2 in the research process can lead to objectual understanding by fulfilling IIC, AC and PEC. However, while AF2 generates valuable predictions that advance objectual understanding in many cases, I will show that it is the researchers’ input that enables the transition from mere predictions to deeper explanatory understanding (EC).

5 Understanding with AlphaFold2

In this section, I analyze how the concepts of objectual and explanatory understanding, as expressed in the conditions IIC, AC, PEC and EC, can be applied to the research surrounding AF2.

AF2 predictions lead to valuable hypotheses on the structures of specific proteins (Sect. 5.1). They also contribute to understanding by generating a large amount of data on protein structures which can lead under some conditions to a deeper understanding of the protein universe (Sect. 5.2). Integrating AF2 into the research process can be interpreted as a new pathway to explanatory understanding (Sect. 5.3).

5.1 Objectual understanding of single proteins with AF2

In one (naïve) way of seeing it, every single AF2-predicted structure could be considered a gain in objectual understanding via data generation counting as fulfilling IIC. This claim is very strong and difficult to evaluate, considering that over 200 mil-

lion structures are available in the AlphaFold Database¹¹ (Varadi et al., 2022), whose overall quality is difficult to assess, as the provided error metrics do not consistently indicate the usefulness of the predictions (see Sect. 4.3).

When researchers investigate the structures of particular proteins, they often do so because they seek a deeper understanding of their function. If a hypothesis constructed using AF2 predictions addresses questions of how a protein could bind to other proteins or molecules based on the structure or why certain parts of the amino acid sequence fold the way they do, the PEC can be fulfilled as well.

This section summarizes successes and limitations of AF2 in understanding the structure of single proteins and evaluates to what extent it fulfills the conditions for understanding.

5.1.1 Successful determination of folded structures with AlphaFold2

Noone et al. (2022) used AF2 in combination with cryo-EM to shed light on a pattern-recognition protein, PTX3, for which no high-resolution structure information was available previously. They began by creating a cryo-EM map of the protein, which did not offer a high enough resolution of its N-terminus. They used AF2 to complement the experimental data and were then able to verify its prediction experimentally. They conclude that “the structural details and functional insights gained from our hybrid cryoEM/AlphaFold full-length structure of PTX3 may contribute to our understanding of these disorders and processes, and aid the rational design of therapies targeting PTX3” (p. 8). It is noteworthy that the researchers explicitly use the term ‘understanding’ in the context of this AF2 prediction. Proteins have specific binding sites formed by their 3D structure that enable selective interactions with other molecules. Based on AF2’s structure prediction, the scientists may be able to identify potential binding sites that could then serve as targets for therapeutic interventions. The use of the AF2-predicted and experimentally validated structures has the potential to fulfill the PEC. Furthermore, the method fulfills the IIC by providing information on a formerly unknown 3D structure, together with cryo-EM and researchers’ analysis.

Terwilliger et al. (2024) analyzed the correspondence between 102 independent cryo-EM maps, AF2 predictions and models deposited in the PDB. They found a mean map-model correlation of 0.56 for AF2 predictions, compared to 0.86 for the deposited models. While they found that the pLDDT was in general a good predictor of the correspondence between experimental data and prediction, 10% of the residues that were predicted with high confidence showed a considerable difference to the data. They conclude that

[E]ven though not all parts of AlphaFold predictions are accurate, they provide plausible hypotheses that can suggest mechanisms of action and allow designing of experiments with specific expected outcomes. Using these predictions as starting hypotheses can also greatly accelerate the process of experimental structure determination. AlphaFold predictions often have very good stereo-

¹¹<https://alphafold.ebi.ac.uk/>

chemical characteristics, making them excellent hypotheses for local structural features. (p. 6)

According to these authors, thus, AF2, in addition to contributing to faster protein structure prediction, clearly fulfills the PEC in suggesting mechanisms of action. For example, AF2 could predict the positioning of substrate-binding pockets in enzymes which gives a suggestion of how catalysis is facilitated. The quote also suggests that AF2 fulfills the AC, as it gives researchers a new way of designing experiments and specifying their expected outcomes.

There is, thus, the potential to gain objectual understanding of single proteins' 3D structure with AF2 via three routes: the predictions can fulfill the IIC by providing new information on proteins, they give researchers a new ability to investigate protein structures (AC) and they can be used to construct hypotheses for mechanisms of action or diseases (PEC). Researchers' expertise is essential to count the conditions as fulfilled.

5.1.2 Limitations of AlphaFold2 in determining folded structures

On the other hand, it is, however, unclear whether each of the over 200 million predictions stored in the AF Database actually can be said to contribute valuable information. The difficulty of assessing the quality of this large number of predictions also introduces extra work for researchers to decide how many of these predictions are valuable for future research.

Alderson et al. (2023) report, for example, that AF2 makes confident predictions for about 15% of human proteins that contain intrinsically disordered regions (IDRs) and conclude that "AlphaFold2 predictions do not reveal functionally relevant structural plasticity within IDRs and cannot offer realistic ensemble representations of conditionally folded IDRs." (p. 1) Cases in which the internal confidence metrics of AF2 are unreliable are arguably particularly problematic, because they might lead researchers to place higher confidence in the predictions than is appropriate. An analysis of the AF2 predictions did, however, reveal that it can be determined based on the pLDDT score whether these proteins with IDRs have a conditional fold (e.g., in the presence of certain binding molecules) or do not have a stable structure at all. They conclude: "Our finding that AlphaFold2 has learned to identify conditionally folded IDRs will enable insights into the sequences, evolution, and structural bioinformatics of these regions" (p. 10). This can be seen as AF2 fulfilling the IIC condition, as its error measure was found to contain information on types of IDRs and can thus be counted as new data available to researchers interested in IDRs. Successfully applied, this also leads to the fulfillment of AC, because it equips researchers with a new method for identifying proteins with conditionally folded IDRs.

Another important point is the above-mentioned conformational variability of certain proteins that cannot be captured in AF2's static predictions. Sala et al. (2023) review approaches that can "increase the structural heterogeneity of the predicted ensemble" (p. 2) by creating a conformationally biased DNN, that is, it predicts 5 output structures with a higher conformational variance. One possibility to achieve this is to use a shallower MSA or (partially) withhold information from templates and

recycling. The correct MSA depth varies from protein to protein, which means that multiple depths have to be tested. For example, Del Alamo et al. (2022) applied this technique to transporter and receptor proteins and report:

Overall, these results demonstrate that both conformations of all eight protein targets could be predicted with AF2 to high accuracy (TM score ≥ 0.9) by using MSAs that are far shallower than the default. However, because the optimal MSA depth and choice of templates varied for each protein, these parameters need to be explored for conformational sampling of a particular target. (p. 3)

This judgment could be made because the conformations had already been determined experimentally. It would be difficult to do if not all conformations were known beforehand, which at the same time would make the investigation less fruitful. They conclude: “while the results presented here provide a blueprint for obtaining AF2 models of alternative conformations, they also argue against an optimal one-size-fits-all approach for sampling the conformational space of every protein with high accuracy” (p. 7) and that their results “highlight the need for further development of artificial intelligence methods capable of learning the conformational flexibility intrinsic to protein structures” (p. 8).

While the AlphaFold Database is thus useful in situations in which researchers need an appropriate template for their investigation, it cannot generally be said that each prediction fulfills the IIC and PEC. Furthermore, the predictions’ usefulness is not always indicated by AF2’s confidence metrics, which complicates the evaluation of the overall quality of the 200 million structures provided.

5.2 Objectual understanding of the protein universe with AlphaFold2

The high number of structures that are freely available in the AlphaFold Database has led to interesting classificatory analyses of the protein universe, which, as researchers argue, can lead to a better understanding of the structure of the protein universe. The protein universe refers to the immense diversity of protein structures and functions. Challenges include understanding the distribution of protein structures within this universe, identifying the reasons for the overrepresentation of certain folds, and characterizing the vast number of unknown or hypothetical proteins. Linking the sequence space to the structure space and identifying evolutionary trajectories across the protein universe are also important research topics.

Bordin et al. (2023) integrated approximately 370,000 of 700,000 structures of 21 model organisms from the AlphaFold database into the CATH database, which “provides an up-to-date and systematic structural classification of protein 3D structures [...]” (Sillitoe et al., 2021, p. 266) by “split[ting] 3D structures into their constituent domains [...] and cluster[ing] these domains into homologous superfamilies where there is sufficient evidence of evolutionary ancestry.” (ibid) The provided information from the additional structures changed the picture of the distribution of proteins in superfamilies, and led them to conclude that “the expansion in structural diversity in CATH superfamilies (i.e., 36% increase in global fold groups) brought by AF2 relatives is exciting, as it could help rationalise functional divergence in these

superfamilies.” (Bordin et al., 2023, p. 10) This is a case in which the condition PEC is fulfilled, as AF2 provided information that led to a potential explanation of the distribution of structures in the protein universe as well as information that could help with characterizing functional divergence. While the role of AF2 is data generation, the generated data are used by scientists to derive potential explanations for the evolutionary development of proteins. It is important to remember that the data quality of this large number of predictions is unclear, which transfers to the potential of explaining the evolution of protein structures with them (see Sect. 5.1.2). It is not possible to achieve explanatory understanding directly from these predictions, but potential explanations derived from them could give researchers new hypotheses to investigate. The objectual understanding they describe also carries the disclaimer ‘According to generally accurate AF2 predictions, this structure distribution seems likely, but the overall quality of predictions remains to be evaluated.’

The researchers also take their study to “[suggest] that the release of a further 214 million models by DeepMind will bring significant insights on how structures (and, therefore their functions) diverge during evolution” (Bordin et al., 2023, p. 2), which sounds like they see a possible contribution to explanatory understanding of structure divergence (EC), as indicated above, via hypothesis generation and experimental verification by scientists. This potential explanatory understanding does not directly relate to the PFP, but would allow a better understanding of a ‘piece of the puzzle’, i.e., the evolution of protein structures and their functions, which could ultimately contribute to solving the PFP, as such data may for example reveal conserved motifs and divergence patterns that could inform the formulation of more general principles.

This research clearly demonstrates AF2’s contribution to the objectual understanding of the protein universe via data generation, while it illustrates at the same time that important contributions from scientists are necessary. The structures to be included had to be selected based on their overall confidence metrics as well as some conditions developed by the researchers, like long stretches without secondary conformations or unusual contact density compared to the PDB. It is thus the background knowledge and interpretative work of the researchers that turns the raw material provided by AF2 into scientific insights.

5.3 Paths to explanatory understanding

As argued in the previous sections, objectual understanding is an interesting result on its own, as it furthers the research process, enriches the available body of information and increases the potential to find new explanations. This section presents a research project in which the objectual understanding gained from AF2 might pave the way for new explanatory understanding.

Mosalaganti et al. (2022) present an in-depth AI-based investigation of human nuclear pore complexes (NPCs). Human NPCs are large multiprotein structures (~1000 proteins) that span the nuclear envelope in the cell, facilitating the selective transport of molecules between the nucleus and the cytoplasm. They are composed of 30 different proteins called nucleoporins (NUPs). Human NPC structure is difficult to analyze experimentally; it comprises many NUPs whose structure could not be determined in sufficient detail with ‘classical’ methods like cryo-EM and X-ray

microscopy, due to low resolution resulting from the conformational dynamics of the pores, the fused inner and outer nuclear membranes, and the presence of intrinsically disordered proteins. Understanding the complex structure of the pores is essential for understanding their function and ultimately the dynamics of nucleocytoplasmic transport. The explanandum thus concerns the dynamic behavior of the pore and to what extent the position and function of the NUPs guide it.

The researchers state that the “[a]nalysis of the structure and dynamics of the NPC at high resolution has been a long-standing goal toward a better molecular understanding of NPC function.” (Mosalaganti et al., 2022, p. 1) They thus explicitly search to further scientific understanding in terms of a functional explanation. With AF2¹², the researchers were able to gain highly accurate predictions of several previously unknown NUPs, which were then validated with X-ray and cryo-EM structures and integrated into additionally “obtained well-resolved cryo-electron tomographic maps of both the constricted and dilated conformational states of the human NPC” (p. 1) to construct a high-resolution model of the NPC. Importantly, all components of the resulting model were thus validated with empirical evidence. The integration of experimental evidence into the model makes this an “observation-backed” context (Boge, 2020, p. 30), which justifies believing that the inferences are relevantly related to the model’s target and provide real explanatory understanding of the concrete dynamics of the real-world pore.¹³ Mosalaganti et al. (2022) state that the resulting model “captures conformational changes that occur during dilation and constriction.” It also “reveals the precise anchoring sites for intrinsically disordered NUPs, the identification of which is a prerequisite for a complete and dynamic model of the NPC” (p. 1). With a combination of AF2 predictions and well-established research techniques that validated them, new mechanistic explanations for both the dynamics of the NPC as well as the positions of intrinsically disordered proteins and their roles in molecular transport were found. This is an impressive result on its own, although it does not touch upon any of the questions posed by the PFP. While AF2 could shed light on a part of the structure with its predictions, allowing the researchers to create a more complete picture of the pore, it also had clear limitations that they propose to solve with different techniques. On the use of AI they conclude:

Not all subunit or domain combinations that we attempted to model with AI-based structure prediction led to structural models that were consistent with complementary data, emphasizing that experimental structure determination will still be required in the future for cases in which a priori knowledge remains sparse. However, even if AI-based modeling does not yield high-confidence results, the models can still serve as tools for hypothesis generation and subsequent experimental validation. (Mosalaganti et al., 2022, p. 7)

¹²They also used predictions from RoseTTAfold, a similar DNN.

¹³This level of confirmation is presumably the reason why Yu et al. (2023) refer to this as “solved scaffold structure“(p. 163).

They thus clearly acknowledge AF2's potential to fulfill PEC and AC. Together with interpretative work of the involved researchers and experimental validation, its predictions enrich the investigation of protein complexes.

This research shows how using AF2 as a method can fulfill all conditions for understanding that I introduced and thereby highlights the role that DNNs can play in the scientific process: first, AF2 predictions were used to gain data on proteins and protein complexes whose structures were unknown before (IIC). It was used to enrich the objectual understanding of the structure of single proteins. Then, these structures were treated as hypotheses and fit into the experimentally determined structure (PEC). The single structure predictions were thus used to further the objectual understanding of the whole NPC by creating a plausible hypothesis of their placement inside the bigger structure, answering the question of how the NUPs are organized spatially within the complex. Based on the greatly enlarged objectual understanding, in a last step, the researchers used the resulting model to create a functional/mechanistic scientific explanation (EC) of the dynamic behavior of the NPC derived from a simulation. Depending on one's position regarding the explanatory character of simulations, the simulation can also be seen as a further hypothesis, but it has been argued (Beisbart, 2012; Boge, 2020; Weisberg, 2013) that the results of simulations can be seen as explanatory. While simulations in themselves cannot guarantee factivity, inferences on the dynamic behavior of the pore seem justified with little uncertainty on these specific explananda, because they are supported by a lot of empirical evidence, both regarding the resulting behavior of the simulated system, with cryo-EM images of the pore in constricted and dilated state, and the AF2-provided NUPs, with additionally experimentally verified NUPs. AF2 also fulfilled AC in providing them with the ability to construct and verify hypotheses on protein complexes using AI predictions. It is, however, important to note that there was a considerable contribution of the researchers on every step of the way.

5.4 A new pathway?

The case study demonstrates that its black box nature does not prevent AF2 from providing objectual understanding in the form of integrating new information and generating hypotheses. Unlike traditional models with well-understood processes, explanations cannot directly be derived from AF2 predictions. With additional effort, however, researchers were able to use the gained objectual understanding and develop explanatory understanding. The process is illustrated in Fig. 3. From step one, the AF2 predictions, which are either predictions of single structures or structure distribution information about many proteins, the conditions can be fulfilled through a second step of either scientific analysis or experimental validation. The AC is directly fulfilled when researchers use AF2 to create new predictions. To arrive at information integration, potential explanations and, in a last step, explanations, thorough scientific analysis is required in addition to the predictions.

I therefore propose that understanding with DNNs can be achieved in a two-step adaptive process from objectual to explanatory understanding. In contrast to more

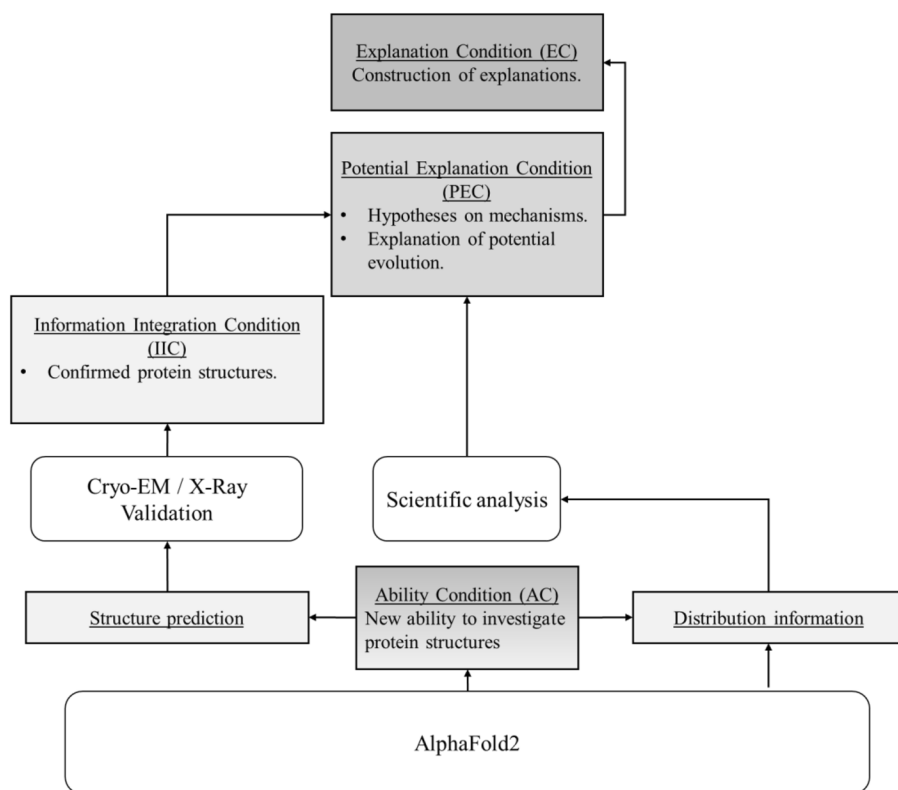


Fig. 3 Two-step adaptive process of gaining scientific understanding with AF2

traditional models, explanations cannot be directly inferred from the results¹⁴ but require an additional step by researchers. Using the DNN in an informed way, keeping track of possibly contained link uncertainty as proposed by Sullivan (2022) and having in mind that its predictions do not equal explanations, as argued by Răz and Beisbart (2022), the research process can profit greatly from the impressive predictions delivered by deep learning. This is in line with the analysis by Agarwal and McShan (2024), who conclude that “[e]ven in the face of an impressive display of accuracy, AF2 is still best used to complement and extend interpretation of experimental data.” (p. 956) Duede (2023) describes the success of AF2 as to “simply bypass computational roadblocks on the way to discovery” (p. 19) and the protein folding expert AlQuraishi (2020) writes in his blog post on AF2 about the possibility of it being a solution, with a strict distinction between the problem of protein structure prediction (cPFP) and the PFP in general:

[T]he core intellectual problem has been solved. [...] In some ways it turned out that protein structure prediction is *easier* than we anticipated. But this is not the point. The point is that before AF2 we didn’t know, and now we do know, that a

¹⁴Though there are exceptions, also for traditional models (Boge, 2021).

solution is possible. My claim is that given this knowledge all the corner cases have become engineering problems. [...] It means that the bulk of the *scientific* problem is solved; what's left now is execution. [...] Protein structure prediction is at the 90% of the problem but 10% of the effort stage.

One item [...] does not fall into this category and ironically it is the one that gave this field its name: the actual dynamic process by which proteins fold. This is a completely different problem and I am almost tempted to say that AF2 has no bearing on it, but that would be far too strong I think – AF2, directly or indirectly, may well contribute to solving the protein folding problem. But that problem remains firmly in the realm of science.

This evaluation corresponds to the results of my investigation and makes clear that while the PFP is not touched, and therefore certainly not solved by AF2, its contribution to the more modest goals of predicting the structure of single proteins, developing hypotheses and equipping researchers with abilities is considerable, and it can generate objectual understanding that could, in the future, aid scientists in solving it.

6 Conclusion

In this article, I argue for the possibility of impressive understanding with DNNs, even without understanding their concrete functioning. I presented four conditions for understanding with a method: information integration (IIC), abilities (AC), potential explanations (PEC), and explanations (EC). In a summary of the challenges that black box models pose for scientific understanding, I argued that the EC cannot be fulfilled by a DNN alone, but always requires researchers to generate an explanation based on the objectual understanding they gained from the DNN's predictions.

As became clear throughout the article, it is very important to specify the object of understanding concisely. Failing to do so might have led to headlines suggesting that AF2 solved the protein folding problem, which is, as I have shown, far from the case. However, AF2 led to progress in several related explananda, like the structure of single proteins, the distribution and evolution of structures in the protein universe, and the dynamic behavior of a protein complex.

The prediction of the structure of single proteins with AF2 leads to the fulfillment of IIC and PEC in many cases, although not generally, due to multiple conformations and intrinsically disordered regions not being reliably predicted by the model. For their fulfillment, it is essential that scientists interpret and verify the predictions. The IIC is then fulfilled by providing more information on the structure of proteins, while the PEC is fulfilled when additional how or why questions are addressed by the results, e.g., how certain parts of the protein behave in terms of binding to other proteins or molecules, or why certain parts of the amino acid sequence fold the way they do.

Furthermore, AF2 fulfills the AC as it equips researchers with new abilities surrounding protein research. Lastly, I have shown how researchers already successfully used AF2 predictions in a two-step adaptive process to gain explanatory understand-

ing, from previously gained objectual understanding, in determining the structure and dynamics of human nuclear pore complexes. While this is far from explaining protein folding and solving the PFP, it is nonetheless a valuable contribution to protein research. All these examples show that while AF2 cannot be seen as a replacement for experimental methods, but rather acts as a (very powerful) data generation method¹⁵, the research process profits greatly from its highly accurate predictions. This is possible because researchers show a profound understanding of the limitations that underlie AF2 predictions and use background knowledge as well as experimental methods to overcome its shortcomings.

While the protein folding problem remains unsolved, it is, to a certain extent, mitigated by AF2 and its extensions. It is now easier for researchers to explore protein structures and their functional implications than ever before. The integration of AF2 into the research workflow has enabled scientists to generate hypotheses about protein behavior and interactions, facilitating a deeper inquiry into biochemical processes. Using AF2's predictions helps researchers prioritize which proteins to study experimentally, thereby optimizing resource allocation and time efficiency in their investigations.

However, it is important to remain aware of the limitations of AF2 and similar deep neural networks. The reliability of predictions can vary depending on the availability of training data and the nature of the proteins being studied. Therefore, researchers must continuously assess the accuracy of AF2 predictions in the context of their specific research questions and be cautious not to over-rely on its predictions. By maintaining a critical perspective and integrating their expert knowledge as well as experimental validation, the scientific community can fully utilize the potential of AF2 and other deep learning models to advance our understanding of phenomena related to protein folding.

With regard to the philosophical background, these results lie between the conclusions of Sullivan (2022) and Răz and Beisbart (2022). While the limitations and unknowns surrounding AF2 severely limit the possible understanding gained from them, as the latter argue, its sensible embedding into the research process led to new objectual and even explanatory understanding, though not in the sense of reducing link uncertainty, but by reasonably embedding AF2 predictions into a two-step adaptive research process, with scientists' background knowledge and empirical verification.

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¹⁵ I thank an anonymous reviewer for pushing me to make this point clearer.

Data availability Not applicable.

Declarations

Conflicts of interest The authors have no relevant financial or non-financial interests to disclose.

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